

The Quotient Topology

Definition 1.0.1 ► Quotient Topology

Let $(X, \mathcal{T})$ and $(Y, \mathcal{T}_1)$ be topological spaces. Then $(Y, \mathcal{T}_1)$ is said to be a **quotient space** of $(X, \mathcal{T})$ if there exists a surjective mapping $f : (X, \mathcal{T}) \rightarrow (Y, \mathcal{T}_1)$ with the property (*).

$$\text{For each subset } U \text{ of } Y, U \in \mathcal{T}_1 \iff f^{-1}(U) \in \mathcal{T}. \quad (*)$$

A surjective mapping f with the property(*) is said to be a **quotient mapping**.

Note.

Trade-off: 陪域上的拓扑越精细, 函数连续越困难; 定义域越精细, 函数连续越简单. 我们为 f “施加压力”, 使得它马上就变得不连续了, 为的就是让连续函数 $g : X \rightarrow Z$ 可以安全且准确地下降为连续函数 $h : Y \rightarrow Z$, 使得 $g = h \circ f$.